Gauss' Theorema Egregium

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

(Dated: October 2026)

Abstract

A complete authored walkthrough of gauss' theorema egregium, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

DG-03 / Differential Geometry / Report v1

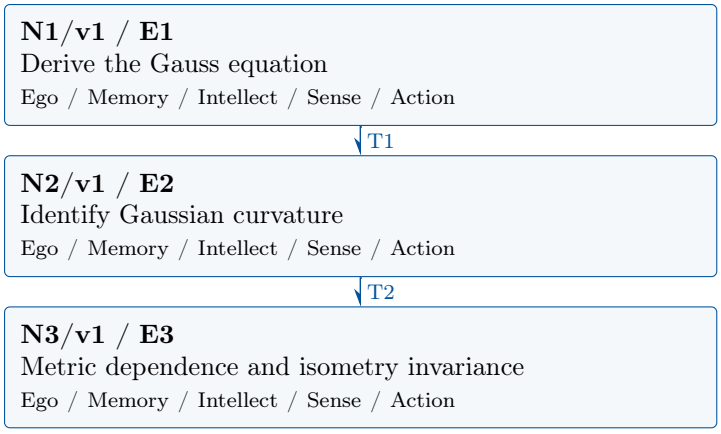
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For a smooth regular surface in $\mathbb{R}^3$, prove Gaussian curvature $K = \det S$ is determined by the first fundamental form and is preserved by local isometries.*

A. Foundations and permitted imports

For a regular surface choose local coordinates and a unit normal ν. Use the induced metric g, its Levi-Civita connection ∇, and decomposition of the ambient derivative $D_X Y = \nabla_X Y + h(X,Y)\nu$ with $D_X \nu = -SX$ and $h(X,Y) = g(SX,Y)$. These follow from orthogonal projection, metric compatibility, and torsion-freeness. Curvature convention $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$. The Gauss equation is derived here.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Intrinsic curvature expressed through h with its exact signs.

Receiving activity. N2: Identify Gaussian curvature

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Orthonormal tangent basis e_1, e_2; prior exports E1/v1

T2: N2/E2/v1 to N3. $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

Receiving activity. N3: Metric dependence and isometry invariance

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Coordinate curvature formula and metric compatibility; prior exports E2/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Derive the Gauss equation

Object and role. Extract tangential terms from ambient curvature zero.

Inputs. Flat ambient connection and tangential/normal decomposition; foundational imports stated in §1

E — Ego

Own E1/v1: Intrinsic curvature expressed through h with its exact signs.

M — Memory

Retain Flat ambient connection and tangential/normal decomposition; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Extract tangential terms from ambient curvature zero.

S — Sense

Changing the curvature convention changes formula signs.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 In R^3, $D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z = 0$.

N1.2 The tangential part of $D_X D_Y Z$ is $\nabla_X \nabla_Y Z - h(Y,Z)SX$; interchange X,Y and subtract.

N1.3 Therefore $R(X,Y)Z = h(Y,Z)SX - h(X,Z)SY$.

N1.4 Pairing with W gives $g(R(X,Y)Z,W) = h(Y,Z)h(X,W) - h(X,Z)h(Y,W)$.

N1.5 This is the Gauss equation with the fixed curvature convention.

Export E1/v1. Intrinsic curvature expressed through h with its exact signs.

Acceptance gate. Changing the curvature convention changes formula signs. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Changing the curvature convention changes formula signs. Recheck dependent nodes N2, N3.

B. N2/v1: Identify Gaussian curvature

Object and role. Compute sectional curvature as a two-by-two determinant.

Inputs. Orthonormal tangent basis e_1, e_2; prior exports E1/v1

E — Ego

Own E2/v1: $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

M — Memory

Retain Orthonormal tangent basis e_1, e_2; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

Compute sectional curvature as a two-by-two determinant.

S — Sense

Reversing ν negates S but preserves its two-dimensional determinant.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Use $X = e_1, Y = e_2, Z = e_2, W = e_1$.

N2.2 The equation gives $g(R(e_1, e_2)e_2, e_1) = h_{11}h_{22} - h_{12}^2 = \det S$, since h is symmetric and the basis orthonormal.

N2.3 For coordinate vectors, $K = g(R(\partial_1, \partial_2)\partial_2, \partial_1)/(g_{11}g_{22} - g_{12}^2)$.

N2.4 The denominator is positive by regularity.

N2.5 Hence Gaussian curvature equals the intrinsic sectional curvature.

Export E2/v1. $K = \det S$ is the scalar computed from the intrinsic curvature tensor.

Acceptance gate. Reversing ν negates S but preserves its two-dimensional determinant. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Reversing ν negates S but preserves its two-dimensional determinant. Recheck dependent nodes N3.

C. N3/v1: Metric dependence and isometry invariance

Object and role. Express the connection and curvature entirely through g.

Inputs. Coordinate curvature formula and metric compatibility; prior exports E2/v1

E — Ego

Own E3/v1: Gaussian curvature is determined by the first fundamental form and preserved by local isometries.

M — Memory

Retain Coordinate curvature formula and metric compatibility; prior exports E2/v1; keep their assumptions and versions attached.

I — Intellect

Express the connection and curvature entirely through g.

S — Sense

Equal extrinsic pictures are unnecessary; the connection is metric-determined.

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 The Christoffel symbols are $\Gamma^k_{ij} = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$.

N3.2 In a coordinate frame $R(\partial_i, \partial_j)\partial_k$ has component $\partial_i \Gamma^\ell_{jk} - \partial_j \Gamma^\ell_{ik} + \Gamma^m_{jk}\Gamma^\ell_{im} - \Gamma^m_{ik}\Gamma^\ell_{jm}$.

N3.3 Thus the displayed K uses only g and its first and second derivatives.

N3.4 A local isometry preserves g and, by uniqueness of the Levi-Civita connection, transports ∇ and R; the scalar K is consequently preserved.

N3.5 This proves the theorem.

Export E3/v1. Gaussian curvature is determined by the first fundamental form and preserved by local isometries.

Acceptance gate. Equal extrinsic pictures are unnecessary; the connection is metric-determined. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Equal extrinsic pictures are unnecessary; the connection is metric-determined. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Derive the Gauss equation

1. In R^3, $D_X D_Y Z - D_Y D_X Z - D_{[X,Y]} Z = 0$.
2. The tangential part of $D_X D_Y Z$ is$\nabla_X \nabla_Y Z - h(Y,Z)SX$; interchange X,Y and subtract.
3. Therefore $R(X,Y)Z = h(Y,Z)SX - h(X,Z)SY$.
4. Pairing with W gives $g(R(X,Y)Z,W) = h(Y,Z)h(X,W) - h(X,Z)h(Y,W)$.
5. This is the Gauss equation with the fixed curvature convention.

B. §4.N2 — Identify Gaussian curvature

1. Use $X = e_1, Y = e_2, Z = e_2, W = e_1$.
2. The equation gives $g(R(e_1,e_2)e_2,e_1) = h_{11}h_{22} - h_{12}^2 = \det S$, since h is symmetric and the basis orthonormal.
3. For coordinate vectors, $K = g(R(\partial_1,\partial_2)\partial_2,\partial_1)/(g_{11}g_{22} - g_{12}^2)$.
4. The denominator is positive by regularity.
5. Hence Gaussian curvature equals the intrinsic sectional curvature.

C. §4.N3 — Metric dependence and isometry invariance

1. The Christoffel symbols are $\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij})$.
2. In a coordinate frame $R(\partial_i,\partial_j)\partial_k$ has component $\partial_i \Gamma_{jk}^\ell - \partial_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell$.
3. Thus the displayed K uses only g and its first and second derivatives.
4. A local isometry preserves g and, by uniqueness of the Levi-Civita connection, transports ∇ and R; the scalar K is consequently preserved.
5. This proves the theorem.

□

V. WORKED ORCHESTRATION

A cylinder of radiusr is parametrized by (r cosu,r sinu,v) and has metric $r^2du^2 + dv^2$. Set $x = ru, y = v$; the metric becomes $dx^2 + dy^2$, so all Γ and R vanish and $K = 0$. Its principal curvatures have one zero value, giving the same detS despite its visible bending.

A. Boundary and hypothesis audit

A sphere has positive K whereas the plane has $K = 0$, so no local isometry can flatten a sphere into the plane. Sense rejects equating arbitrary bending with an isometry; the metric, not only the embedding, must be preserved.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A sphere has positive K whereas the plane has $K = 0$, so no local isometry can flatten a sphere into the plane. Sense rejects equating arbitrary bending with an isometry; the metric, not only the embedding, must be preserved.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard Math 230a: Riemannian geometry, curvature, and Hopf–Rinow.